

Assignment for Interviewee

Objective:

Build a prototype application that demonstrates Text-to-RAG (Retrieval-Augmented Generation) capabilities using Google Cloud Platform (GCP) services, Vertex AI, and a Streamlit front-end, applied on publicly available financial data.

Scope & Requirements

1. Data Acquisition (Mandatory)

- You are required to **search and crawl/scrape public data sources** to collect financial data.
- Focus on **P&L / Income Statements** and **Balance Sheets** of **publicly listed companies**.
- Acceptable public sources include (examples only, choose any):
 - SEC 10-K/10-Q filings
(<https://www.sec.gov/edgar/searchedgar/companysearch>)
 - Yahoo Finance / Google Finance
 - Company Investor Relations pages (annual reports, earnings reports)
- Collected data should include (at minimum):
 - **Income Statement:** Revenue, Gross Profit, Operating Income, Net Income, EPS.
 - **Balance Sheet:** Assets, Liabilities, Shareholder Equity, Cash & Cash Equivalents.

(Note: You are not allowed to use private APIs or paid datasets. Only publicly available financial data is expected.)

2. RAG Pipeline

- Preprocess the collected data into a suitable format for retrieval.
- Build a **retrieval layer** (embeddings + vector search or equivalent).
- Use **Vertex AI** for embeddings and/or model inference.
- The pipeline should support **natural language queries** and respond with context-backed answers.

3. Stream lit Application

- Input box for a user to ask financial questions.
 - Display of the model's generated answer.
 - Display of retrieved supporting context (e.g., P&L section snippet).
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4. Deployment & GCP Usage

- Use GCP services where applicable (Vertex AI, Cloud Storage, Big Query, etc.).
 - Local execution is acceptable, but documentation must describe how you would deploy the solution on GCP.
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Deliverables

1. **Source Code** (GitHub/Zip).
 2. **Documentation (README)** including:
 - High-level architecture diagram.
 - Data acquisition approach (how and from where you crawled/scraped).
 - List of GCP services used and their role.
 - Setup instructions.
 - Sample queries with screenshots and video run of outputs.
 - Limitations and possible improvements.
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Constraints

- No pre-provided dataset: you must **collect financial data yourself**.
 - Keep the prototype simple but functional.
 - Avoid using external paid APIs outside GCP.
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Sample Natural Language Queries to Support

- *"What was Tesla's net income in 2022 compared to 2021?"*

- *“How much cash did Apple report on its balance sheet in FY2023?”*
- *“Which company had a higher EPS in 2022, Microsoft or Google?”*
- *“What percentage of assets are financed by liabilities for Amazon in 2021?”*
- *“Compare operating income of Tesla across the last three years.”*